

Network protocol

In our Network Protocols there are many security flaws which affect the system. As new technologies evolve new landscapes of vulnerability and assault will increase. Additionally, these security flaws are not distinct; they can be found in other protocols as well. To shield from unknown and known threats we need to stay up-to-date of security measures. In general, we have so many security payload flaws which affect the network, the following are a few examples:

Note: RFC means Request for change number which means it's a set of instructions which tells how data flows in the internet. In precise way we can say that if we want to know how exactly data flows into the internet, network, security layers.

1. Internet Protocol: - The IP contains a set of instructions which perform data transmission over the network called as datagrams. The IP is more responsible for packetizing the data and sending it to the destination from the source. The data should be fragmented and have properties of reassembling the data. There are two versions of IP: IPV4, IPV6.

RFC: 791 (Internet Protocol, 1981)

Security Flaws: IP does not provide any form of encryption or authentication, due to which it is vulnerable to man-in-the-middle attacks. To mitigate this, we are using additional security protocols i.e., internet protocol security which provides encryption and authentication mechanism and removes the reliability and integrity of data transmitted via the internet. (Hall & Comer, 2015)

2. Secure Shell (SSH):- it is a network protocol used to securely connect to remote devices in un-disciplinary networks. This is mostly used for remote machine to execute commands but supports tunneling, forwarding TCP ports and x11 connections.

The SSH has few versions i.e., SSH-1, SSH-2.

SSH-1: RFC: 4250, It has several security flaws in key exchange algorithms which made attackers easy to gain access of private key which is used for authentication. Due to flaws the SSH-1 is deprecated. (S. & Ed, 2006)

SSH-2: It has several RFCs on protocols, authentication, transport layer, connection protocols they are RFC-4251, RFC-4252, RFC-4253, RFC4254 respectively (T. & Ed, 2006). The vulnerability in SSH-2 was caused by CRC32

algorithm used to detect the tampering. This vulnerable was patched by updating the affected versions of SSH server (National Vulnerability Database, 2008).

References: -

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